Simulating Visual Impairments in a CAVE Environment: A Symptom-Based Approach

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***Abstract*— This paper presents the design, implementation, and evaluation of a comprehensive visual impairment simulator developed for the Immersive 3D Visualization Lab (CAVE environment). The primary objective of the system is to enable architects and designers to evaluate the accessibility of spaces by directly experiencing perceptual limitations. Utilizing a symptom-based approach rather than relying solely on clinical diagnoses, the system employs custom shaders to simulate specific visual symptoms such as blurred vision, visual field loss, and color vision deficiencies. The simulation is embedded within realistic scenarios of daily living, including meal preparation and personal hygiene, performed in a single-family house model. The effectiveness of the simulator was verified through a study involving 51 participants, divided into control and experimental groups. Results indicated that the simulated impairments significantly increased task completion times and perceived difficulty, validating the system's impact on user performance. Participants rated the realism of the simulated disorders at 3.96 (±0.92) and the environment at 4.1 (±0.78), confirming the tool's potential for educational purposes and functional diagnostics in inclusive design.**

Keywords*— Virtual Reality (VR), CAVE System, Visual Impairment Simulation, Inclusive Design, Accessibility, Symptom-Based Simulation, Human-Computer Interaction

I. Introduction

Modern virtual reality (VR) technologies are finding increasingly broad applications, extending beyond entertainment to play crucial roles in medicine, education, and functional diagnostics. The capacity for immersive reality reproduction renders VR a unique tool that allows healthy individuals to experience the world from the perspective of those with visual impairments. This is particularly significant in the context of architectural design, where understanding end-user limitations is vital for creating accessible and inclusive environments (inclusive design).

This article presents the design and implementation of a comprehensive visual impairment simulator within the Immersive 3D Visualization Lab (LZWP) environment, as well as an analysis of the results of the conducted study regarding the effectiveness and usability of the simulator. The primary objective of the system is to enable architects, designers, and students to evaluate the accessibility of designed spaces through the direct experience of perceptual limitations.

In contrast to models based solely on clinical diagnoses, the proposed system utilizes a symptom-based approach. This allowed for the implementation of simulations for diverse conditions, including refractive errors, retinal and optic nerve diseases, and color vision deficiencies. The simulation operates in real-time using custom shaders and post-processing techniques, which allows for smooth adjustment of symptom intensity and their arbitrary combination. Test scenarios are embedded in the context of activities of daily living, such as meal preparation or operating devices, carried out within a single-family house model.

II. Theoretical Background

A. Literature overview

To identify the current state of knowledge regarding vision impairment simulation in virtual environments, a systematic literature review was conducted. The analysis encompassed databases such as IEEE Xplore, Scopus, Web of Science, ACM Digital Library, and PubMed. The review focused on studies published after 2020, addressing simulation methods, technologies employed, and the validation of VR/AR solutions in the context of visual impairments.

1) Simulation Methods and Objectives in Existing Research

The analysis of the selected studies indicates two main trends in the application of virtual reality: immersive simulations and diagnostic tools. In the context of immersive simulations, researchers focus on replicating difficulties in daily functioning for the purposes of education or behavioral analysis. For instance, the Seeing other perspectives

(OpenVisSim) [1] project utilizes both VR and AR to simulate glaucoma, demonstrating that visual field loss in the inferior region has a more critical impact on task performance than loss in the superior region. A similar research approach is presented in the article The worse eye revisited: Evaluating the impact of asymmetric peripheral vision loss on everyday function [2]. The authors employed a gaze-contingent dynamic blur simulation on healthy subjects, proving that asymmetric peripheral vision loss significantly increases reaction time and necessitates increased head movements. In turn, the VIVR system integrates a VR headset with a physical controller in the form of a white cane and a vibration system, which enhances the sense of presence and enables spatial orientation training for the blind.

A significant area of research involves the use of VR as a tool to support diagnostics and functional assessment of patients. In the study Use of Virtual Reality Simulation to Identify Vision-Related Disability in Patients With Glaucoma, a simulation of daily living tasks was employed without digitally generating visual impairments to identify the degree of disability in actual patients. The findings confirmed that user performance in the virtual environment correlates with standard clinical measures of visual function. Concurrently, research is being conducted to verify VR as a screening tool. The article Virtual reality-based and conventional visual field examination comparison in healthy and glaucoma patients compared visual field examinations using VR headsets with traditional perimetry, while maintaining full procedural standardization. It was demonstrated that VR can serve as an effective method for glaucoma identification, yielding results comparable to conventional techniques. The broader potential of these technologies is described in Artificial intelligence and digital solutions for myopia, indicating that VR is applicable not only in detecting visual field defects but also in the assessment and treatment of other conditions, such as strabismus and amblyopia.

2) *Technologies and Programming Tools*

The majority of the analyzed solutions rely on the Unity3D engine, which has established itself as the standard for developing applications of this nature. Advanced shaders (e.g., OpenGL) and real-time image post-processing techniques are utilized to generate visual effects that simulate visual impairments, such as peripheral blur or visual field loss. In terms of hardware, Head-Mounted Display (HMD) solutions predominate, exemplified by devices such as the HTC Vive, Oculus Quest, and Fove VR. Identification of the Research Gap

Despite the widespread application of VR headsets (HMDs) in the literature, there is a lack of solutions dedicated to Cave systems. Existing studies primarily focus on the interaction of a single individual with the environment via HMDs. Furthermore, the majority of the analyzed simulators concentrate on a narrow spectrum of conditions, predominantly glaucoma. There is a demand for a comprehensive tool simulating a broader range of visual impairments that could support spatial designers in creating accessible environments (inclusive design). The present project aims to address this gap through the implementation of a simulator in the Immersive 3D Visualization Lab (LZWP).

B. *Scope of Visual Impairments Considered*

The proposed simulator is designed to reproduce perceptual effects corresponding to a set of sixteen selected visual impairments and ocular diseases. Instead of modeling individual clinical diagnoses in isolation, the system follows a symptom-based approach, reflecting the fact that many visual conditions exhibit overlapping visual effects. The selection of impairments is based on clinically documented visual symptoms observed in refractive, retinal, optic nerve, lens-related, and functional vision disorders. Accordingly, the impairments are grouped into three categories, as described below.

1) *Refractive and Optical System Impairments*

The first group of impairments comprises conditions related to the optical properties of the eye and image formation mechanisms. Refractive errors such as myopia, hyperopia, and presbyopia primarily affect visual acuity as a function of viewing distance, resulting in blurred perception of distant or near objects. Astigmatism introduces directional blur and geometric distortions, commonly perceived as bending or doubling of straight lines. Lens-related disorders, including cataract, further degrade image quality through global haze, reduced contrast, glare, halo effects around light sources, and characteristic color shifts.

2) *Retinal, Optic Nerve and Visual Field Disorders*

The second group includes impairments affecting the retina, optic nerve, and the integrity of the visual field. Age-related macular degeneration (AMD) primarily impacts central vision, leading to central scotomas, reduced acuity, image distortion, and decreased contrast sensitivity. Diabetic retinopathy is associated with fluctuating visual clarity, dark spots, and floating opacities, whereas retinal detachment may cause flashes of light, shadow-like curtains intruding into the visual field and progressive peripheral vision loss. Glaucoma is characterized by gradual constriction of the visual field, often progressing toward tunnel vision.

3) *Functional and Perceptual Visual Disturbances*

The third group encompasses functional impairments and perceptual disturbances that significantly affect visual experience without being confined to a single anatomical structure. This category includes color vision deficiencies (protanopia, deuteranopia, tritanopia), which involve selective attenuation of specific color channels. It also covers adaptation-related conditions such as nyctalopia (night blindness) and hemeralopia (day blindness), characterized by impaired vision under conditions of low or high luminance, respectively. Additionally, this group includes perceptual phenomena such as diplopia, nystagmus, and vitreous floaters, which manifest as image duplication, oscillatory visual motion, and the presence of moving opacities within the visual field.

III. SYSTEM DESIGN AND IMPLEMENTATION

A. *Assumptions*

1) *General Objectives and Intended Use of the System*

The project assumed the development of a simulator of visual impairments and eye diseases intended to operate in an immersive virtual reality cave environment. The main objective of the system was to enable healthy individuals to experience visual perception in a manner as close as possible to the subjective way in which people with various visual impairments perceive the world.

The system has an educational and research-oriented character, and its primary application was to support the education of students in fields such as architecture and urban planning, as well as to enable architects and designers to evaluate the accessibility of designed spaces from the perspective of people with visual impairments.

2) Methodological Assumptions of the Simulation

A symptom-based approach to the simulation of visual impairments was adopted, consisting in the reproduction of characteristic perceptual effects (symptoms). The simulation has an approximate nature, and its main assumption is to allow users to experience visual limitations in the context of everyday activities and movement within space. It was assumed that multiple symptoms could be combined simultaneously and that their intensity could be smoothly adjusted, allowing for the representation of a wide spectrum of visual experiences.

3) Simulated Visual Impairments and Eye Diseases

The project assumed the simulation of visual perception corresponding to selected visual impairments and eye diseases, represented by sets of characteristic visual symptoms. This scope included both refractive errors and disorders of the retina, optic nerve, as well as functional vision disturbances.

Within the system, the simulation of the following visual impairments and eye diseases was implemented: myopia, hyperopia, presbyopia, astigmatism, cataract, glaucoma, age-related macular degeneration, diabetic retinopathy, retinal detachment, Benson's disease, nystagmus, diplopia, night blindness, hemeralopia, vitreous floaters, and color vision deficiencies (color blindness).

Each of the listed impairments was implemented in the system as a predefined set of symptoms, while users were able to utilize both predefined configurations and custom combinations of perceptual effects through arbitrary symptom selection and intensity adjustment.

4) Technical and Organizational Environment

The system was designed to operate in the BigCAVE environment located in the Immersive 3D Visualization Lab (hereinafter referred to as LZWP) at Gdańsk University of Technology. At any given time, the system was used by a single user whose position and head orientation were tracked using sensors installed in the room. These sensors registered the movement of stereoscopic glasses equipped with dedicated markers, allowing the displayed image to be dynamically adjusted to the user's position.

Interaction with the virtual environment was carried out using a dedicated handheld controller (flystick). In addition, a simulation operator was present alongside the user and was responsible for managing the system state and the parameters of the simulated visual impairments.

5) Scenarios and Space

The spatial scope of the simulation included a model of a single-family house interior at a 1:1 scale, consisting of several functional rooms. This environment was assumed to provide a neutral and commonly recognizable context for the implementation of scenarios based on activities of daily living. The system supported the execution of simple user tasks, such as meal preparation, performing hygiene-related activities, or searching for objects. These tasks served as tools for assessing the impact of visual impairments on everyday functioning within a domestic environment.

6) Performance and Ergonomic Assumptions

The project assumed that the simulator should ensure a high level of user comfort under immersive virtual reality conditions. Particular emphasis was placed on system smoothness, image stability, and the spatial coherence of visual effects in order to reduce the risk of discomfort or simulator sickness. The user interface and interaction methods were assumed to be intuitive and as simple as possible, enabling the use of the system by individuals without prior experience with virtual reality technology. Management of simulation parameters was implemented as an external operator function, which helped maintain a high level of immersion. The system was required to be compatible with the infrastructure of the LZWP and designed in a manner that allowed for further expansion.

B. Design

The design of the visual impairment simulator was based on a set of basic functional and research assumptions. The system was developed in an iterative manner, where design decisions were gradually refined during development and verified through regular testing in the virtual reality cave environment. Observations from test sessions and early pilot use were used to adjust both system behavior and interaction rules.

The system was designed to run in a CAVE environment and to be used by a single participant inside the projection space. The participant interacts only with the virtual environment, while all simulation settings are controlled by the system operator. These settings include the type and strength of visual impairments, the active room, and the task sequence. This separation of roles helps keep experimental conditions consistent between sessions.

Visual impairments are simulated during simple daily tasks rather than isolated perception tests. The virtual environment represents the interior of a residential building at a 1:1 scale and consists of several rooms with different functions. The room layout and object placement support tasks based on object search, navigation, and basic object manipulation. The layout was adjusted to the physical size of the cave and the available tracking area.

A symptom-based approach was used instead of modeling specific eye diseases. Individual visual effects represent common perceptual symptoms and can be activated separately or combined. The intensity of each effect can be adjusted, which allows flexible configuration of simulation conditions for different test scenarios.

The user interface was designed in a diegetic manner and integrated directly into the virtual environment. Interaction elements are presented as part of the scene rather than as external menus or overlays. Despite the overall simplicity of the interface, the system supports a wide range of interactions to increase immersion. Users can interact with multiple objects and environment elements using natural actions such as pointing, selecting, manipulating, and activating objects.

During later design iterations, changes were introduced based on functional tests and pilot studies. These changes mainly concerned room layout, task structure, object placement, and general interaction rules. The iterative design process allowed the system to be gradually refined without changes to its overall structure.

C. Implementation

1) Sceneries/Rooms

The simulator consists of 7 scenes in total. Two of these (Main and Setup) act as holders for the core control elements. Both scenes feature an identical SimulationSetup object (maintained via DontDestroyOnLoad()), which serves as the heart of the simulation. It includes the following modules:

- GameManager: Manages logic and command processing.
- SoundManager: Controls the audio system.
- VisualImpairmentsManager: Handles visual impairment effects.
- SocketManager: Manages data exchange between the simulator and the external controller.
- LZWPlib Package: Includes cameras, tracker glasses, hand controllers, and the projection logic for the 3D CAVE environment.

The primary difference lies in the loading method. The Setup scene loads new environments in Single mode, unloading previous ones. The Main scene utilizes a SceneriesManager that additively loads the entire MainCollection at startup, then toggles between environments by activating or deactivating them. The Main scene is the default for standard use, while Setup remains a fallback for operators and testing purposes. The remaining 5 scenes are the user-facing environments, referred to as "sceneries":

- Intro: a 3D lobby/main menu designed to familiarize the user with the simulation and present project information.
- Scenario Sceneries (Hall, Bathroom, Kitchen, Bedroom): these are arranged in an inverted "T" layout. The Hall serves as the central hub, connecting to the Bedroom (left), Bathroom (straight ahead), and Kitchen (right).

Hall: the initial entry point for test scenarios, and connector between other sceneries. It features an "illusory space" to make the environment optically larger (to balance otherwise not very house-like structure, and increase immersion) and three sets of doors acting as triggers to load the other rooms.

Fig. 1 Hall scenery - central hub connecting all test environments.

Bathroom: equipped with standard fixtures (sink, toilet, shower, storage) and a door leading back to the Hall. Balance between the amount of movement space and static points of interest.

Fig. 2 Bathroom scenery - environment designed for hygiene-related tasks.

Kitchen: features a refrigerator, stove, cabinets, small appliances, workspace, and a return door to the Hall. This room has plenty of movement space, which is perfect for search/exploratory tasks.

Fig. 3 Kitchen scenery - environment optimized for object search and sequential tasks.

Bedroom: a room with the lowest amount of movement space - designed for more static tasks. Furnished with a bed, big wardrobe, TV, and bedside lighting. Same as two previous, this scenery also has a door leading to the Hall.

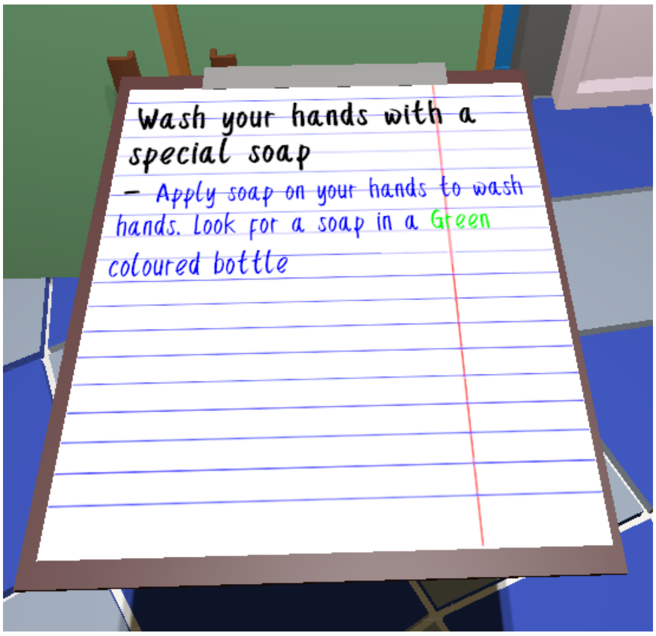
Fig. 4 Figure X. Bedroom scenery - compact layout intended for static interaction tasks.

All four sceneries were built to match CAVE size (3,4m x 3,4m) with different colors, structure, and a wide range of interactive elements (exceeding those required by the basic test tasks) to enhance immersion. Additionally, each environment includes interactive light sources, such as ceiling lamps.

2) Test Scenarios

As part of the experimental procedure, participants execute scenarios based on activities of daily living, such as meal preparation, personal hygiene, and operating consumer electronics. To precisely monitor user progress, a Task Tracking System was implemented.

The user interface was designed in a diegetic form as an interactive 3D object placed within the scene, which the participant can pick up and manipulate in virtual space. This object serves as a dynamic task log. The content display mechanism relies on sequential filtering: the system presents only those subtasks required at the current stage of the simulation. Additionally, visual coding of the task status was applied: the active task is highlighted with a bold line, while successfully completed stages are marked with a strikethrough, providing the user with immediate feedback.



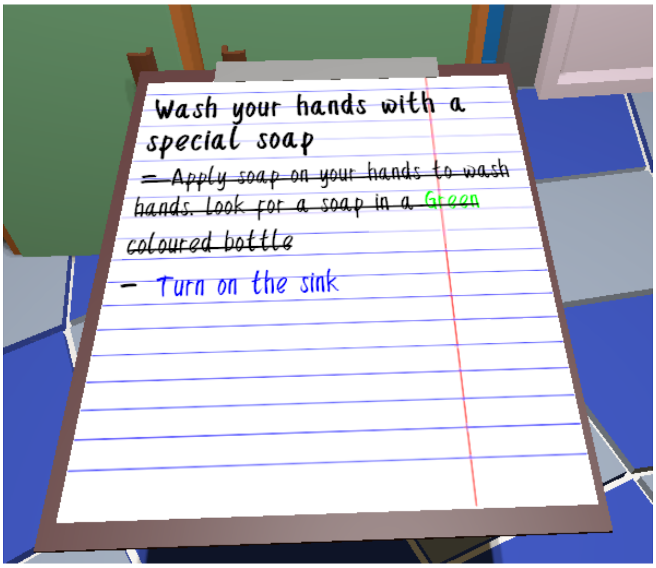


Fig. 5 Task tracking interface: (a) before completing a subtask; (b) after subtask completion with visual status update.

The test scenarios consist of tasks requiring the participant to utilize various motor-perceptual abilities, such as manual precision, visual search, and hand-eye coordination. The technical specification of the individual tasks is presented below:

a) Meal preparation (Sandwich Assembly): This task constitutes a simulation of a sequential process. The user is required to assemble the target object (a sandwich) by manipulating ingredients in accordance with strict instructions. The system enforces adherence to a specific sequence of actions, whereby any attempt to utilize an incorrect ingredient is either blocked or results in no progress. Upon the successful conclusion of the sequence, the individual elements are merged into a single interactive object, allowing for subsequent manipulation (e.g., picking up the finished sandwich).

b) Personal hygiene (Hand Washing): This task is divided into several stages requiring observational skills and motor precision. In the first instance, the participant must identify and locate a specific soap object within the scene, which is distinguished by a predefined color. The subsequent step involves turning on the faucet to activate the virtual water stream. The actual handwashing action is simulated by placing one or both controllers (flystick) directly under the water stream, contingent upon the number of controllers currently being actively tracked by the CAVE system. The system validates this activity once the hands are maintained within the stream for a specified duration. The task concludes upon re-interacting with the faucet to turn off the water flow.

c) Consumer electronics operation (TV Activation): This scenario comprises two key subtasks: tool retrieval and device operation. The participant initiates the task by locating the television remote control within the room and retrieving it. Subsequently, utilizing the remote, the user must activate the television and cycle through channels until the specific channel indicated in the task instructions is located. As feedback, the television displays a static noise effect accompanied by a text overlay indicating the current channel number. The system validates the task as complete once the user selects the correct channel and maintains this state for a defined verification duration.

3) Simulator external controller

A significant portion of the simulator, including visual impairment filters, test scenarios, user positioning, and application diagnostics is managed by an external controller application. Developed in C# using the WPF framework, this application functions as a server for connected simulator instances. It operates over a local network (LAN), allowing the operator to control the simulation from a separate workstation.

Communication Workflow:

- Discovery: Upon startup, the server initiates a broadcast via UDP.
- Connection Establishment: The main simulator instance listens for this broadcast, extracts the server's IP address and port, and establishes a connection for itself and all other instances using the TCP protocol. TCP is utilized to ensure data integrity and reliable packet delivery.
- Data Synchronization: Once the connection is established, updates from the controller are transmitted in real-time to all simulator instances, where they are processed by the relevant management scripts.

Controller Interface Layout:

- Connected Apps: Located at the top, showing currently connected clients and their IPs in the local network.
- Network Configuration: With the Ip Address, Broadcast Port and Communication Port fields responsible for configuring the connection.
- Sphere Renderer Transform: A section dedicated to managing the transform properties of the SphereRenderer.
- Visual Impairments: A dedicated panel for controlling active filters, including the ability to manage and load predefined presets.
- Dev Console: A command-line interface for manual overrides and specific triggers, such as starting a scenario or forcing a scene change.

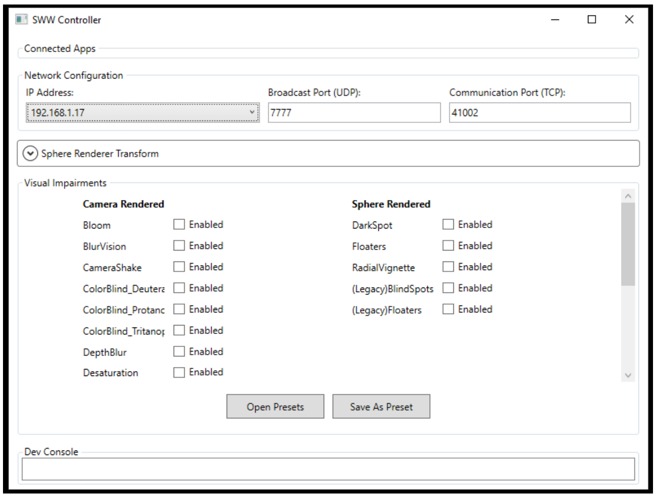


Fig. 6 External controller application interface.

4) *Shader-Based Simulation of Visual Impairments*

To simulate selected visual impairments in the immersive CAVE environment, a set of custom shaders was developed using the Unity engine. The shader-based approach was chosen due to its high performance, flexibility, and suitability for real-time applications in large-scale immersive systems. Each impairment effect is implemented as a modular filter, making it possible to combine several visual effects dynamically during runtime.

Depth-aware blur effects were used to simulate conditions such as farsightedness [3]. These shaders use the camera depth buffer to calculate the distance of visible objects from the viewer. Based on this distance, blur intensity is smoothly adjusted using focus distance and focus range parameters. This creates a gradual transition between sharp and blurred areas, which reflects depth-related vision problems. To achieve a stable and visually smooth image, the blur implementation uses multi-sample filtering with rotated sampling patterns. This approach reduces artifacts such as double vision or directional streaks, which are especially noticeable in immersive environments. The implementation of the depth-aware blur is demonstrated in Figure 7.

Fig. 7 Simulation of farsightedness using a depth-aware blur algorithm.

Other visual impairments, including image distortion, line deformation, and motion-related discomfort effects (e.g., dizziness or camera shake), are implemented using spatial transformations based on view direction or world-space position rather than screen-space coordinates [4]. This ensures that the effects remain spatially coherent across all projection planes of the CAVE system. The resulting distortion effects are illustrated in Figure 8.

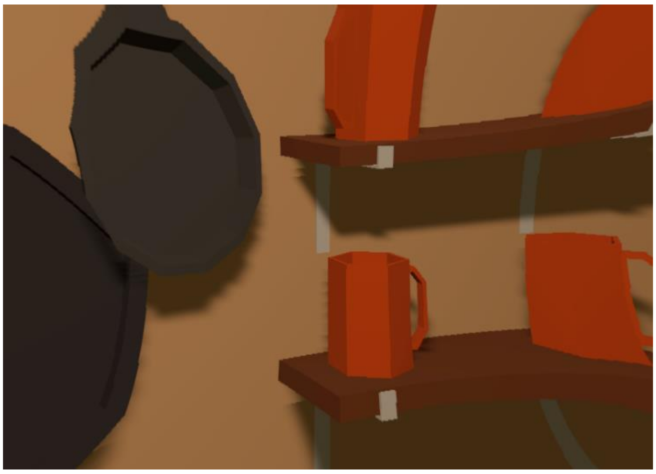


Fig. 8 Simulation of visual distortion effect.

Some of the implemented shaders operate directly on the image's color representation in order to simulate color perception impairments. These effects use per-pixel color transformations, such as linear color space conversions and channel mixing, which approximate conditions such as red-green color vision deficiencies (protanopia and deuteranopia), blue-yellow deficiency (tritanopia) as well as general reduced color sensitivity [5]. Figure 9 illustrates the effect of the color transformation matrix.

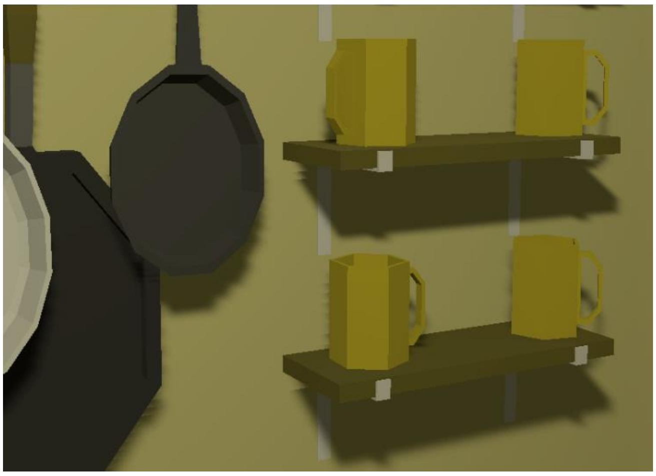


Fig. 9 Simulation of protanopia achieved through per-pixel color matrix transformation.

All implemented shader-based effects share a common parameter that controls the intensity of the simulated impairment. This parameter allows smooth adjustment of effect strength, from normal vision to increasingly strong visual distortions. Using a unified intensity control helps maintain a consistent user experience and simplifies experimental setup when comparing different types of visual impairments.

5) *Interaction with the Environment*

Interaction with the virtual environment is a key component of the developed VR-based visual impairment simulator, as it enables users to naturally explore and manipulate objects within the scene. The application is controlled in real time using 3D glasses and up to two handheld controllers (flysticks), one for each hand. Head movements are directly mapped to camera motion, ensuring that the virtual camera continuously follows the user's gaze, which enhances immersion and spatial awareness.

The handheld controllers are used to point at virtual objects and initiate interactions. Each controller emits a virtual ray with a fixed length of approximately two meters, allowing the user to select objects within reachable distance. When the ray intersects with an interactive object, visual feedback is provided by highlighting the object, indicating that it is ready for interaction. This mechanism allows users to precisely target elements of the environment even in complex scenes.

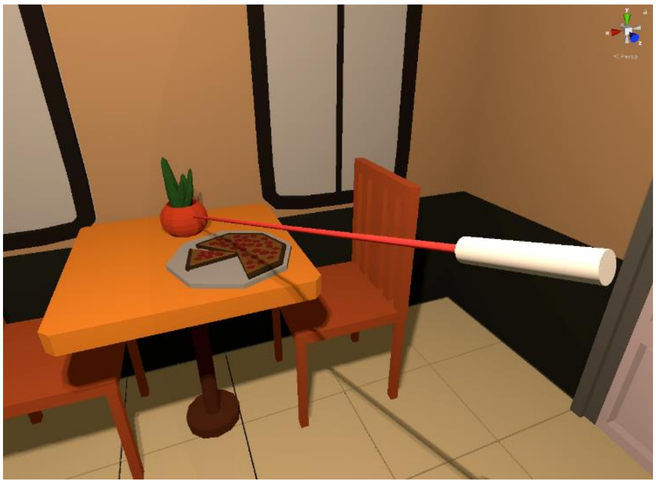


Fig. 10 Flystick object visualization.

Fig. 11 Ray-based object selection: (a) object outside interaction range; (b) object detected and highlighted.

The interaction system is based on a limited set of physical buttons available on the controllers. Although the hardware provides multiple input options, the current implementation uses three buttons to ensure simplicity and usability. The primary interaction button ("Fire") is responsible for direct actions such as opening or closing cabinet doors, toggling light switches, and picking up objects. This button represents the main mode of interaction and is used most frequently during the simulation.

Fig. 12 Holding an interactable object in the air.

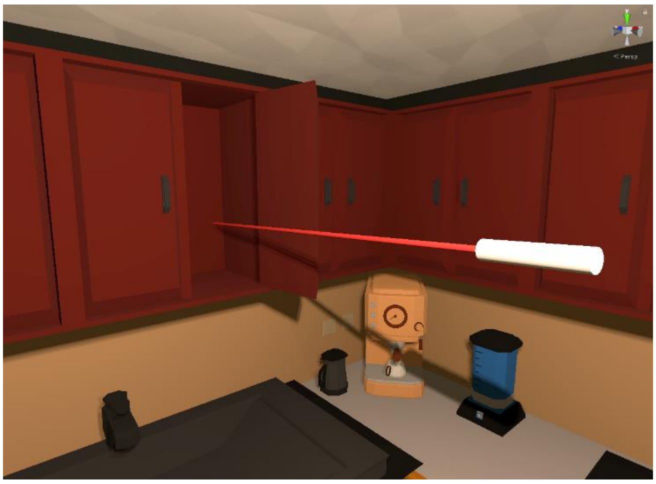


Fig. 13 Cabinet door interaction: (a) closed state before activation; (b) opened state after activation.

A secondary interaction button ("Button1") enables context-dependent actions. For example, when the user is holding an object using the primary button, pressing the secondary button allows interaction with elements of that object, such as pressing a button on a remote control to turn on a television. This design allows for more complex interaction scenarios without increasing the cognitive load on the user.

An additional button ("Button4") is used to toggle the visibility of the controller's aiming aids, including the interaction ray and the visual representation of the controller itself. Although this function does not directly manipulate objects, it serves as an important part of the user interface, allowing users to adjust the level of visual assistance according to their preferences and current task.

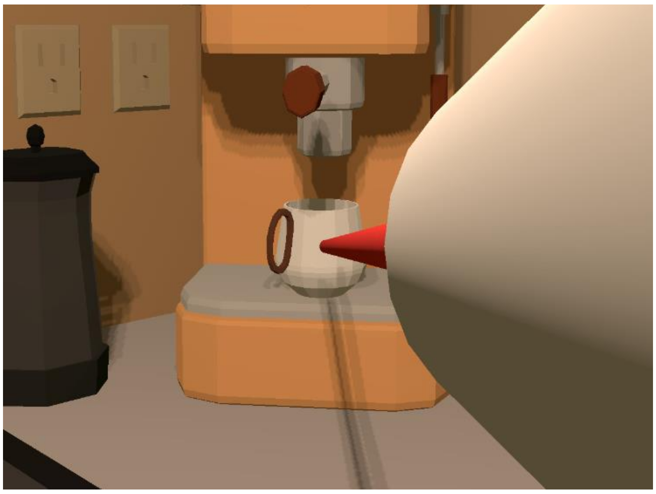


Fig. 14 Flystick visibility toggle: (a) controller and ray visible; (b) controller hidden while interaction remains active.

The interaction design is closely aligned with the specific requirements of the CAVE virtual environment and the functionality provided by the LZWPLib library. The use of flysticks and ray-based selection was largely dictated by the hardware and software constraints of the platform, as well as by commonly adopted VR interaction standards. The system deliberately supports a limited set of interaction types, including object selection, object manipulation, and click-based actions performed while holding objects. These interaction types were chosen as they represent the most natural and intuitive forms of user interaction within immersive virtual environments and are sufficient to support all scenarios required by the simulator. As a result, the implemented interaction system provides an intuitive and reliable means of interacting with the virtual environment, while maintaining simplicity and usability.

IV. Research Methodology

A. Research Questions and Hypotheses

The research conducted within the framework of the project focused on evaluating the effectiveness and usability of the developed simulator of visual impairments and eye diseases in an immersive virtual reality environment. The main area of analysis was to verify whether the applied simulation mechanisms enable a realistic representation of perceptual limitations characteristic of selected visual impairments and whether this experience is perceptible to users who do not have visual deficits.

It was assumed that the VR application is capable of faithfully reproducing key perceptual symptoms associated with individual visual impairments, which should lead to a noticeable deterioration of visual perception in healthy users. It was expected that the use of the simulator would result in subjectively perceived perceptual difficulties, confirming the effectiveness of the adopted symptom-based approach.

An important aspect of the research was also the analysis of the impact of simulated visual limitations on the performance of typical activities of daily living in a virtual environment. It was assumed that the presence of simulated visual impairments would significantly increase the subjective level of difficulty of performing user tasks, such as navigating the space or manipulating objects, compared to conditions without active simulation.

In parallel, the potential of the application as a tool supporting the architectural design process was analyzed. The hypothesis was adopted that the possibility of directly experiencing perceptual limitations in an immersive virtual environment may constitute valuable support for architects and designers, enabling the evaluation of accessibility and functionality of designed spaces from the perspective of people with visual impairments.

The formulated hypotheses formed the basis for defining the research procedure and selecting methods of subjective assessment, aimed at a comprehensive verification of both the realism of the simulation and its potential practical applications.

B. Research Group

The study included 51 participants aged 18 years and older. The research group consisted of volunteers who participated in experimental testing of the proposed VR-based visual impairment simulator.

All participants met the basic inclusion criteria related to the safe use of immersive virtual reality systems. These criteria included the absence of medical contraindications to VR environments, such as photosensitive epilepsy, motion sickness, or balance disorders, normal or corrected-to-normal vision (e.g., use of corrective glasses or contact lenses), and the ability to comfortably perceive stereoscopic content.

Exclusion criteria included a history of severe motion sickness or simulator sickness, diagnosed photosensitive epilepsy, and severe uncorrected visual impairments that could interfere with VR perception or safe participation in the experiment.

Previous experience with immersive VR systems was not required. However, participants' familiarity with VR technology was recorded, as it could influence task performance and subjective perception of the simulated visual impairments. This factor was taken into account during the analysis to distinguish between the effects caused by the simulated vision disorders and those related to user experience with immersive systems.

Participation in the study was voluntary, and all participants were informed about the purpose and procedure of the experiment prior to participation.

C. Research Methods and Tools

The study was conducted as a laboratory experiment in a virtual reality cave environment. The goal was to collect quantitative and qualitative data related to the perception of simulated visual impairments during simple tasks performed in a virtual space.

Participants completed a set of tasks based on activities of daily living in a virtual residential environment. The tasks were performed under different visual impairment simulation settings, including a mode without active visual effects. During each session, task completion time and the number of errors were recorded automatically by the simulator.

After completing the task-based part, participants filled in a questionnaire evaluating the perceived realism of the simulation and the difficulty of the tasks. A five-point Likert scale was used, along with short open-ended questions.

In addition, a short semi-structured interview was conducted after each session. The interview was used to collect general feedback on system behavior and perceived issues related to perception and interaction.

For the study, a basic set of research tools was prepared, including a session scenario, a consent form, a questionnaire, and an interview guide. A pilot study was conducted before the main experiment to verify the study flow and introduce minor organizational adjustments.

D. Variables and Threats to Validity

The study employs a defined set of variables to evaluate the realism and usability of a virtual reality visual impairment simulator. The independent variables include the type of simulated visual impairment, its severity, and possible combinations of visual conditions. All participants complete the same predefined test scenario composed of multiple tasks, which remains fixed across experimental conditions. The dependent variables comprise objective performance measures, such as task completion time and error rate, as well as subjective ratings of perceived realism and task difficulty collected using a five-point Likert scale. To support internal validity, relevant confounding and latent factors were considered, including prior virtual reality experience, age, fatigue and learning effects, spatial abilities, susceptibility to cybersickness, and unrecognized minor vision defects.

Construct validity may be constrained by the simulator's limited ability to fully represent the subjective and complex nature of real-world visual impairments. Potential internal validity threats, such as learning effects and visual fatigue, are acknowledged, although the fixed experimental scenario and short study duration are expected to limit their impact. External validity may be limited by the use of a sample predominantly composed of students and university staff, while the study design allows for the inclusion of participants with diagnosed visual impairments in a validation group. Conclusion validity may be affected by the sensitivity of the measurement instruments and may be addressed through pilot testing, questionnaire refinement, and the combination of subjective assessments with objective performance metrics and system logs.

V. STUDY EXECUTION AND RESULTS ANALYSIS

A. Study Execution

1) Study Participants

The study included 51 people (16 women and 35 men) aged 18 to 45. The largest group was participants aged 18–25 (78% of the sample). Participants were recruited from students and employees of Gdańsk University of Technology.

To take part in the study, participants had to have normal vision or vision corrected to normal with glasses or contact lenses. This requirement was checked before the study procedure began.

Two groups were identified in the study:

- Control group - 8 people who performed tasks without any active simulati
- Experimental group - 43 people who performed tasks with a simulation of selected vision disorders enabled.on of visual impairment.

To take part in the study, participants had to have normal vision or vision corrected to normal with glasses or contact lenses. This requirement was checked before the study procedure began.

2) Study procedure

The study was conducted in the Immersive 3D Visualization Lab at Gdańsk University of Technology. The participant stood inside a projection space where three walls, the floor, and the ceiling displayed images synchronized with the user's position. The motion-tracking system used passive markers placed on protective glasses and cameras, allowing the perspective to adjust in real time to the position of the user's head. The total duration of one session was 15–20 minutes.

The procedure was as follows:

- Participants were introduced to the purpose of the study, informed that the session would be recorded, and asked to give informed consent for data processing.
- A short training session on how to use the immersive visualization system was provided.
- The participant performed tasks in a prepared virtual environment representing a home. Each participant completed the following three tasks in a fixed order:
    - Washing hands in the bathroom - find the indicated soap of the correct color (green), turn on the tap, place the controllers under running water for a specified time, and turn off the tap.
    - Preparing a sandwich in the kitchen - find the ingredients and arrange them in the correct order according to the on-screen instructions.
    - Turning on the TV in the bedroom - locate the remote control, turn on the TV, and find the indicated channel by switching programs.

  Each participant performed these three tasks under one previously assigned simulation of a visual impairment. During the study, the time needed to complete each task was measured.
- A subjective questionnaire was conducted in which participants rated task difficulty, level of visual discomfort, and overall user experience.
- A final interview was conducted to collect opinions and suggestions about interaction, usability difficulties, and possible improvements to the application.

3) Simulated Vision Disorders

- The study included eight types of simulated eye diseases and vision problems:
- Benson's disease
- Glaucoma
- Night blindness
- Diabetic retinopathy
- Nystagmus
- Retinal detachment
- Cataract
- Macular degeneration

The selected conditions included disorders affecting visual acuity, color perception, contrast, and field of vision. This made it possible to analyze different perceptual limitations while performing everyday tasks in a virtual environment.

B. Results analysis

1) Average Task Completion Time

The average time needed to complete each task was calculated for both the control group and the experimental group. The results are shown in the Table 1 and illustrated in a comparison chart of the two groups in Figure 15.

TABLE 1 Comparison of the average task completion time for the control group and the experimental group

	Task 1 – Bathroom	**Task 2 – Kitchen**	**Task 3 – Bedroom**
Control group	1 min 41 s	2 min 6 s	1 min
Experimental group	2 min 48 s	2 min 14 s	1 min 55 s

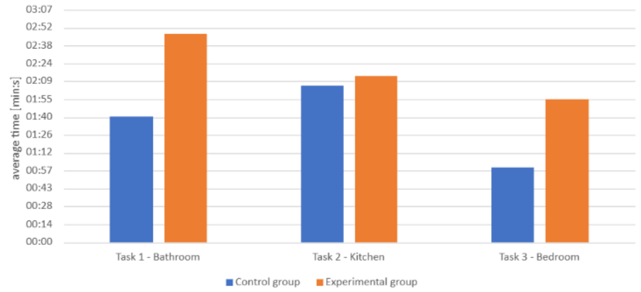


Fig. 15 Comparison of the average task completion time for the control group and the experimental group.

In all analyzed tasks, the experimental group - performing activities with simulated vision problems - needed more time than the control group.

The largest time difference was observed in Task 1 (bathroom), where the average completion time in the experimental group was longer by 1 minute and 6 seconds. A clear increase was also seen in Task 3 (bedroom), with a difference of 55 seconds. The smallest difference occurred in Task 2 (kitchen), at 8 seconds, but the experimental group still performed the task more slowly.

These results confirm that simulated vision disorders increased the time needed to perform everyday activities in the virtual environment. However, the size of this effect depended on the type of task.

2) Detailed Comparison of Simulated Vision Disorders

To better analyze how each simulated vision problem affected task performance, the average completion times were calculated separately for all eight simulations. This approach makes it possible to identify which vision disorders caused the greatest increase in task time and whether the type of difficulty depended on the task.

The average time to complete each task for the eight vision disorders is shown in the Table 2.

TABLE 2 Average Task Completion Time

		Average Task Completion Time		
No.	***Vision Disorder***	***Task 1 – Bathroom***	***Task 2 – Kitchen***	***Task 3 – Bedroom***
1	Benson's disease	2 min 51 s	2 min 44 s	1 min 52 s
2	Glaucoma	3 min 28 s	2 min 04 s	2 min 19 s
3	Night blindness	2 min 55 s	2 min 20 s	3 min 23 s
4	Diabetic retinopathy	3 min 34 s	2 min 33 s	2 min 04 s
5	Nystagmus	2 min 19 s	2 min 10 s	1 min 49 s
6	Retinal detachment	1 min 45 s	1 min 51 s	56 s
7	Cataract	3 min 40 s	2 min 33 s	1 min 36 s
8	Macular degeneration	1 min 58 s	1 min 27 s	1 min 25 s

The results were also presented in a chart in Figure 16, including values from the control group to allow direct comparison.

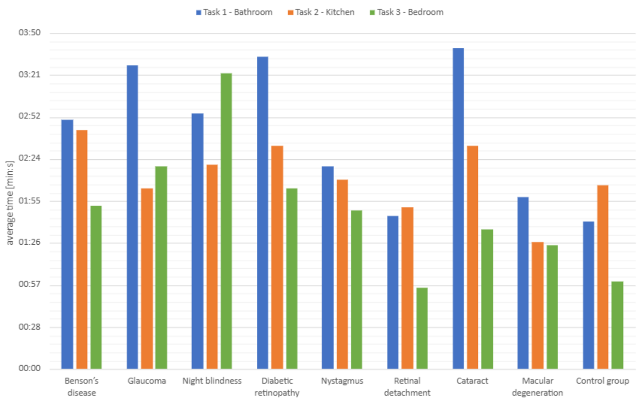


Fig. 16 Comparison of the average task completion under a selected vision disorder.

Task completion time varied depending on the type of simulated vision disorder. The longest times were observed for cataract, diabetic retinopathy, and glaucoma, especially in Task 1 (bathroom). Night blindness caused the greatest delay in Task 3 (bedroom), which is consistent with the nature of this condition and difficulties in low-light environments.

Cataract and diabetic retinopathy clearly slowed tasks that required good visual sharpness and precise recognition of details. The shortest times were observed for retinal detachment and macular degeneration, where perceptual limitations had the smallest effect on performance

The control group achieved shorter and more consistent completion times in all analyzed rooms. Differences between the groups were especially visible in Task 1 and in the bedroom task under night-blindness simulation.

The results indicate that the type of simulated vision disorder had a significant impact on task performance. The greatest difficulties were caused by conditions that reduce visual sharpness, contrast, and the ability to function in low lighting.

In addition, the potential influence of participants' age on task performance was examined. The table below presents the average completion time for each task depending on the age group: 18-25 years (34 participants) and 26-65 years (9 participants) within the entire study group.

TABLE 3 Comparison of the average task completion time by age group

Age	**Task 1 – Bathroom**	**Task 2 – Kitchen**	**Task 3 – Bedroom**
18-25	2 min 31 s	1 min 59 s	1 min 42 s
26-65	3 min 51 s	3 min 11 s	2 min 49 s

The average task completion time depending on age indicates that participants aged 26-65 performed all tasks noticeably slower than those aged 18-25. The largest difference was observed in Task 1 (bathroom), where the average completion time was 3 min 51 s for the older group and 2 min 31 s for the younger group. A similar tendency was observed in Task 2 (kitchen) and Task 3 (bedroom). These results may suggest that participants' age had some influence on the pace of task execution in the virtual environment.

However, it should be noted that the group sizes were unequal: the 18-25 age group included 34 participants, while the 26-65 group consisted of only 9 participants. Therefore, the obtained averages should be treated as an approximate comparison between the groups. To reliably assess the impact of age on task performance, further studies with more balanced group sizes would be required.

3) Difficulty Assessment and Impact of Simulated Disorders

The assessment of task difficulty and the impact of the simulated disorders was based on responses given by participants in the final questionnaire.

The overall difficulty rating for performing tasks under simulated vision disorders was 3.27 (±1.2) on a five-point Likert scale. This result indicates a moderate level of difficulty experienced by participants while completing tasks in the virtual environment. It means that participants did feel the limitations caused by the simulations. The vision disorders affected how the tasks were performed and the speed of work, but they were not strong enough to make the tasks impossible to complete.

The chart in Figure 17 shows the percentage distribution of difficulty ratings for each task based on the final questionnaire. The data illustrate participants' subjective assessment of task difficulty in the three analyzed rooms: bathroom, kitchen, and bedroom.

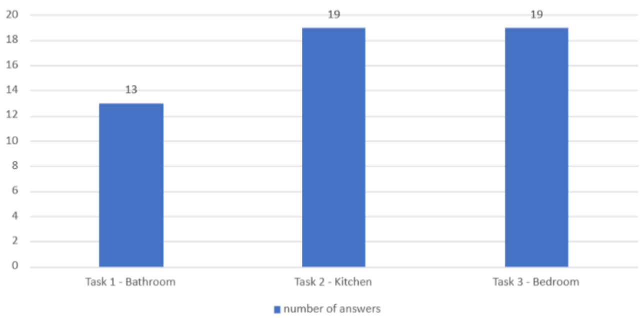


Fig. 17 Assessment of task difficulty

Detailed analysis of responses showed that Task 2 - preparing a sandwich in the kitchen (37.25% of responses) and

Task 3 - turning on the TV and finding the selected channel in the bedroom (37.25%) were considered the most difficult. Task 1 - activities in the bathroom - was less often indicated as the most difficult (25.49%).

For Task 2, the difficulty was mainly due to the need to find several ingredients placed in different parts of the room and to perform a complex, multi-step sequence of actions. This required both good spatial orientation and precise identification of objects.

For Task 3, the main difficulty factor was the reduced lighting in the bedroom. These conditions, combined with the simulated vision problems, limited the visibility of surrounding elements, which increased both the completion time and the perceived difficulty.

4) *Realism of the Simulator*

Participants rated the realism of the virtual environment and the simulated vision disorders on a five-point Likert scale. The average scores are shown in the chart in Figure 18.

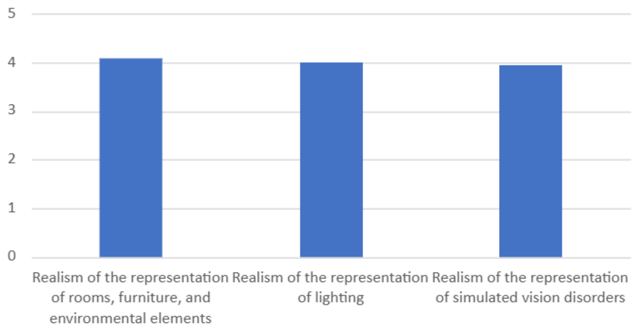


Fig. 18 Assessment of realism of the Simulator

The realism of the rooms, furniture, and environmental elements was rated 4.1 (±0.78). This indicates an overall high evaluation of the virtual environment. The highest scores were given to the rooms themselves and surrounding elements, which was also confirmed by many positive comments from participants. They highlighted the aesthetics and graphic quality, mentioning "nice graphics," "clear compared to other simulations," and a "well-designed environment." Some participants also described the project as visually attractive, calling it an "interesting visualization" with a "retro atmosphere," which further increased the sense of immersion.

The realism of lighting was also rated highly at 4.0 (±0.83), showing a positive reception of this part of the simulation. However, in written comments participants noted difficulties in darker areas of the scene. The bedroom and the inside of the refrigerator were especially problematic, where visibility was clearly limited. The issues with the refrigerator were not due to design errors but to technical limitations of the environment (the CAVE room could not be completely closed). For this reason, tasks were intentionally designed so that using the refrigerator was not required. As a result, some products were placed in unusual locations, which may have seemed less intuitive but was a deliberate solution to technical constraints.

Participants emphasized that "the lighting made tasks much harder," and dark objects placed against a dark background were difficult to recognize. On one hand, these observations may show that the simulation effectively reproduced visual limitations and perceptual difficulties. On the other hand, they point to areas that could be improved, especially contrast and lighting of key scene elements.

The realism of the simulated vision disorders was rated 3.96 (±0.92), indicating a generally positive evaluation but with more varied opinions.

Some participants stated that the simulations convincingly showed the difficulties experienced by people with vision impairments. They noted that the experience encouraged reflection on everyday challenges, for example: "after a few minutes, a simple task becomes incredibly problematic." This suggests the application has strong potential for building empathy and increasing awareness in design. At the same time, a few comments questioned certain visual effects, such as "I don't know any eye condition that causes waves in vision." These remarks may indicate a need for further refinement of some simulations to better match the typical symptoms of specific vision disorders.

It is worth noting that some participants reported having personal experience with a given vision problem (now corrected or treated) and confirmed that the visual effects matched their real perceptual experiences. These opinions support the validity of the simulations and suggest that, despite some doubts, many of the representations were consistent with the subjective experiences of people affected by these vision impairments.

5) *Qualitative Analysis of Observed Difficulties*

During the study, various difficulties were observed, and their nature depended on the type of simulated vision disorder. These problems were identified both from participants' statements and from researchers' observations during task performance. Most often, they concerned reading text, recognizing colors, identifying objects, and functioning in low-light conditions.

Problems with reading instructions and task lists occurred especially in the simulations of Benson's disease and cataract, where participants reported blurred vision and reduced sharpness. In glaucoma and diabetic retinopathy simulations, the main difficulties involved distinguishing colors and finding objects with similar colors, which increased task completion time. Night blindness clearly made it harder to function in the poorly lit bedroom, while in the case of nystagmus, occasional reading problems were reported due to image instability.

The simulations of retinal detachment and macular degeneration had the smallest impact on comfort and task efficiency. In these cases, difficulties were rare and occasional. Overall, the reported problems were consistent with the characteristics of the specific disorders, which confirms the accuracy of the applied simulation effects.

VI. Conclusion

1) *Answers to the research questions and verification of hypotheses*

The study made it possible to evaluate the realism and usefulness of an application that simulates vision disorders in a virtual reality environment. The results show that the simulator effectively reproduces perceptual limitations typical for the analyzed conditions. This is confirmed by the average realism rating of simulated vision disorders at 3.96 ± 0.92, as well as by the longer task completion times in the experimental group compared to the control group.

High ratings for the realism of rooms (4.1 ± 0.78) and lighting (4.0 ± 0.83) indicate a positive reception of the virtual environment and visual effects. Reported difficulties - especially with reading text, recognizing details and colors, and functioning in low-light conditions - confirm that the simulation realistically reflected specific vision problems. Even participants without vision impairments experienced clear perceptual limitations while using the application, including difficulties with spatial orientation, longer task times, and higher subjective difficulty ratings.

Based on the analysis, the first three research hypotheses were confirmed. The application realistically reproduces vision limitations characteristic of selected disorders, and the presence of simulated conditions significantly increased task completion time compared to the control group. The average difficulty rating of 3.27 (±1.2) indicates a noticeable perceptual load on participants.

Regarding the fourth hypothesis, the application shows potential as a tool supporting architectural design. Experiencing the simulation helps users better understand problems caused by vision deficits, which may support the design of more accessible and functional spaces.

2) *Practical significance of the project*

The results suggest that the simulator can serve both educational and design purposes. In particular, it highlights the importance of proper lighting, contrast, text readability, and spatial arrangement of elements.

User experiences show that even simple activities, such as preparing a sandwich or finding an object, can become significantly more difficult with certain vision disorders. These conclusions may support the design of spaces adapted to the needs of people with visual impairments.

3) *Study limitations and directions for future research*

Some limitations of the study should be noted. The sample included 51 participants, mostly aged 18–25, which may limit the generalization of results to other age groups. It would be especially valuable to include older participants (up to 65 years old), who are more likely to experience many of the analyzed vision disorders, particularly degenerative ones.

Future studies could involve a larger and more diverse sample, including people with real vision impairments. This would allow verification of the realism of the simulations and comparison with real-life experiences. Due to methodological and organizational constraints, the study included only selected vision disorders. In the future, it would be useful to include other common conditions, such as myopia (nearsightedness) and hyperopia (farsightedness), to provide a broader evaluation of the application's functionality and usefulness.

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